

1.) a. Constraint 1: Add artificial a_1
(=)

$$\rightarrow x_1 - x_2 + x_3 + a_1 = 2$$

Constraint 2: Add slack s_1
($\leq$)

$$\rightarrow x_1 + x_2 + 2x_3 + s_1 = 10$$

Constraint 3: Subtract surplus s_2 , add artificial a_2
($\geq$)

$$\rightarrow x_1 + x_2 - x_3 - s_2 + a_2 = 6$$

b. $\min w = a_1 + a_2 \quad == \quad \max w' = -(a_1 + a_2)$

Phase 1 standard form LP:

$$\text{Max } w' = -a_1 - a_2$$

s.t.

$$x_1 - x_2 + x_3 + a_1 = 2$$

$$x_1 + x_2 + 2x_3 + s_1 = 10$$

$$x_1 + x_2 - x_3 - s_2 + a_2 = 6$$

$$x_1, x_2, x_3, s_1, s_2, a_1, a_2 \geq 0$$

c. Simplex Tableau:

	x_1	x_2	x_3	s_1	s_2	a_1	a_2	RHS
a_1	1	-1	1	0	0	1	0	2
s_1	1	1	2	1	0	0	0	10
a_2	1	1	-1	0	-1	0	1	6
w'	-2	0	0	0	1	-1	-1	-8

Using x_1 as pivot ($w' = -2$)
min ratio

$$a_1: \frac{2}{1} = 2$$

$$a_2: \frac{6}{1} = 6$$

$$s_1: 10$$

$\Rightarrow a_1$ leaves basis

Pivot on $x_1 = 1$ in $R1$:

Tabikan $\rightarrow$	x_1	x_2	x_3	s_1	s_2	a_1	a_2	RHS
$R1:$	x_1	1	-1	1	0	0	1	2
$R2:$	s_1	1	1	2	1	0	0	10
$R3:$	a_2	1	1	-1	0	-1	0	6
$w':$	w'	-2	0	0	0	1	-1	-8

$$R2 \leftarrow R2 - R1$$

$$\rightarrow R2: [0, 2, 1, 1, 0, -1, 0 | 8]$$

$$R3 \leftarrow R3 - R1$$

$$\rightarrow R3: [0, 2, -2, 0, -1, -1, 1 | 4]$$

$$w' \leftarrow w' + 2R1$$

$$\rightarrow w': [0, -2, 2, 0, 1, 1, -1 | -4]$$

Using x_2 as pivot ($w' = -2$)
min ratio

$$s_1: \frac{8}{2} = 4$$

$$a_2: \frac{1}{2} = 2 \rightarrow a_2 \text{ leaves basis}$$

2.) a. Constraint 1: Subtract Surplus S_1 , add artificial a_1
($\geq$) $\rightarrow x_1 - x_2 - S_1 + a_1 = 8$

Constraint 2: Add Slack S_2
($\leq$) $\rightarrow 2x_1 + 3x_2 + S_2 = 24$

Constraint 3: Add Slack S_3
($\leq$) $\rightarrow 2x_1 + x_2 + S_3 = 12$

Standard Form: Max $z = 2x_1 + x_2$
 $s.t.$

$$x_1 - x_2 - S_1 + a_1 = 8$$

$$2x_1 + 3x_2 + S_2 = 24$$

$$2x_1 + x_2 + S_3 = 12$$

$$x_1, x_2, S_1, S_2, S_3, a_1 \geq 0$$